



DD ROBOCON 2026

AUTODESK FUSION AWARD - GENERATIVE DESIGN CHALLENGE

OPTIMIZE  CONSOLIDATE  PERFORM

[LINK TO REGISTER](#) 

SUBMISSION DEADLINE

24th JUNE 2026 23:00 India Time

(submission link will be shared after registration)

Calling all DD Robocon 2026 teams! Participate and win exciting awards..

Autodesk Fusion Generative Design Challenge 2026 invites participating teams to showcase engineering optimization and manufacturing-driven design using Autodesk Fusion.

Modern competition robots require lightweight structures, efficient assemblies, and rapid manufacturing. Generative Design enables teams to explore optimized solutions that can improve robot performance while reducing weight, complexity, and manufacturing effort.

This challenge is directly aligned with DD Robocon robot development activities and enables teams to apply AI & Generative Design workflows on components and assemblies already being developed for competition robots.

Participants will demonstrate lightweight engineering, assembly optimization, manufacturability, and performance improvement using real robotics applications.

Learn more about Autodesk Fusion: <https://www.autodesk.com/products/fusion-360>

Autodesk Learning: <https://www.autodesk.com/learn>

Please read participation guidelines and terms & conditions before submission.

AWARDS

MAIN AWARDS - Eligible ONLY for teams attending DD Robocon event

1st Prize	INR 25,000
2nd Prize	INR 15,000
3rd Prize	INR 10,000

RECOGNITION AWARDS

Promising Innovation Award	INR 5,000
Best Engineering Application Award	INR 5,000
Best Manufacturable Design Award	INR 5,000

COMPETITION STRUCTURE

Each participating team shall complete 2 Generative Design tasks. BOTH tasks have to be completed and submitted as entry. Submissions must be based on components or assemblies belonging to the team's actual DD Robocon 2026 robot design.

TASK 1 – LIGHTWEIGHT OPTIMIZATION

Teams shall select one existing robot component and optimize it using Generative Design.

Original Robot Component → GD Study → Optimized Design → Final Submission

Suggested applications could be Manipulator arms, Gripper supports, Wheel hubs, Launcher brackets, Structural members, End effectors, Drive supports or any other component. These are only listed as examples, you are free to choose any other component also.

You will need to following for making the final PPT	Mandatory comparison metrics
• Original component design • Final optimized GD part • Fusion public link • Render images • Design description	• Original mass • Optimized mass • Percentage weight reduction

TASK 2 – PART CONSOLIDATION

Teams shall select one existing robot assembly and redesign it into an integrated single part solution using Generative Design. Minimum requirement is assembly with 3 or more original parts that are consolidated into one.

Multi-Part Assembly → GD Study → Integrated Design → Final Submission

Suggested applications could be End effector systems, Sensor mounting assemblies, Manipulator joints, Drive supports, Structural interfaces, Robotic subassemblies or any other component. These are only listed as examples, you are free to choose any other component also.

You will need to following for making the final PPT	Mandatory comparison metrics
• Original assembly • Consolidated design • Fusion public link • Render images • Design description	• Original part count • Final part count

SUBMISSION REQUIREMENTS

Each team shall submit 1 PPT file with precisely 6 slides:

<u>Slide # 1 for INTRO</u>	Institute Name Team Name DD Robocon ID Team Photo
<u>Slide # 2 and 3 for Task 1 Lightweight Optimization</u>	Original component image Optimized GD component image Mass comparison Design explanation
<u>Slide # 4 and 5 for Task 2 Part Consolidation</u>	Original assembly image



	Consolidated part image Part count comparison Design explanation
<u>Slide # 6 for Fusion Links</u>	Link 1- Fusion public link for Lightweight Optimization entry Link 2- Fusion public link for Part Consolidation entry Link 3- Fusion public link for Robot 1 Link 4- Fusion public link for Robot 2

FAQ's

1. Can teams use the same subsystem for both exercises?

Yes. Teams may use the same subsystem provided both exercises demonstrate different engineering objectives.

2. Is physical fabrication mandatory?

No. Submission focuses on design and Generative Design workflows. ONLY Fusion GD to be used.

3. Can concept models be submitted?

No. Entries must originate from actual DD Robocon robot development.

4. Are both Tasks to be done?

Yes. Both tasks are part of the competition. Strictly follow the given PPT guidelines.

5. What designs are not allowed?

Generic CAD practice models, Internet sourced designs, Non-Robocon components, Pure aesthetic topology studies, Unrelated concept parts, Designs not linked to actual competition robots

6. How do I get more help or information?

Attend the Fusion Workshop series (online) to interact with us and know more. We will help you with all questions and technical assistance also.



FUSION WORKSHOP SERIES

We will be organizing a series of online sessions to help you through this competition.

Workshop 1 - 10th June 2026 (4:30 pm to 5:30 pm)

Topic: Competition Launch & Submission Guidelines

Video call link: <https://meet.google.com/wih-bnpz-avm>

Workshop 2 - 17th June 2026 (4:30 pm to 5:30 pm)

Topic: AI and Generative Design for Robotics Applications

Video call link: <https://meet.google.com/zke-hdco-wmf>

Detailed Topics: Lightweight Optimization

- Component selection
- Preserve geometry setup
- Constraints and load definition
- Design optimization workflow
- Robotics examples

Part Consolidation

- Assembly identification
- Consolidation methods
- Integrated design strategies
- Manufacturing considerations
- Robocon applications

For any queries/questions, please email us at

autodesk-partner@curiosity3d.com